FROM RESEARCH TO INDUSTRY

cea tech

SYSTEM MODELING INTRODUCTION

papyrus

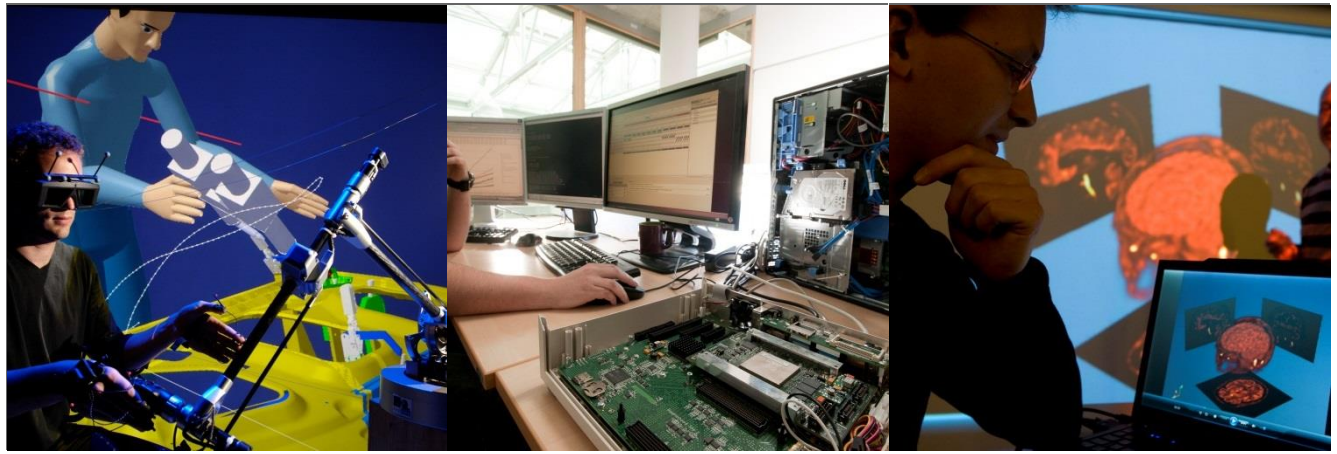
www.eclipse.org/papyrus

list

Introduction on UML for Industrial Systems

François Terrier, Shuai Li, Jérémie Tatibouët, Sébastien Gérard,
Ansgar Radermacher, Asma Smaoui {first_name}.{last_name}@cea.fr

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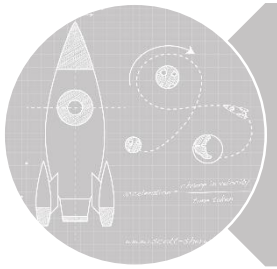
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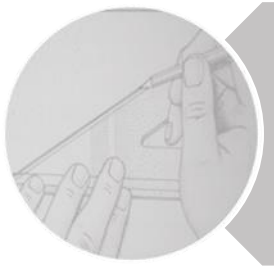
Course material is are available on https://github.com/asmasmaoui/INSTN_B2



Introduction



Basis is Modeling



Modeling with Which Language?

What is a System?

➔ **Complex** and **heterogeneous** systems
responding to real-world events



Human interactions





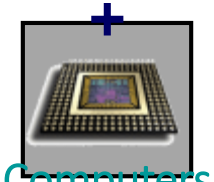
Physical interactions
sensors / actuators

Embedded system



Software

+



Computers

+



Networks





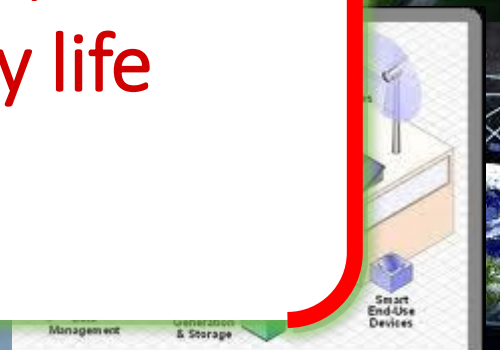


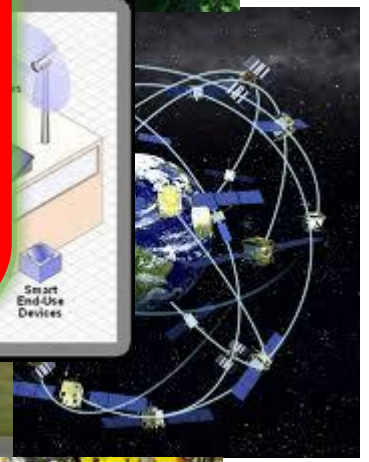
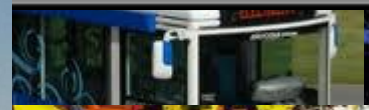
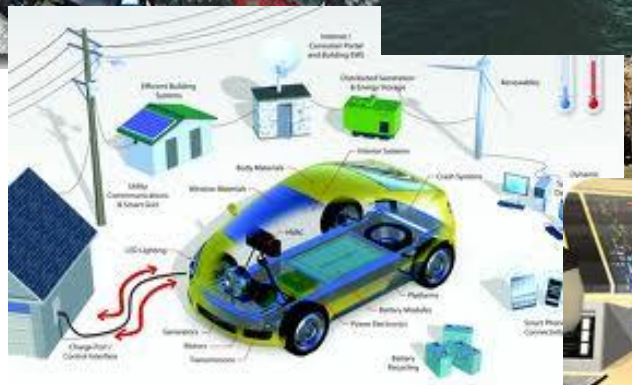










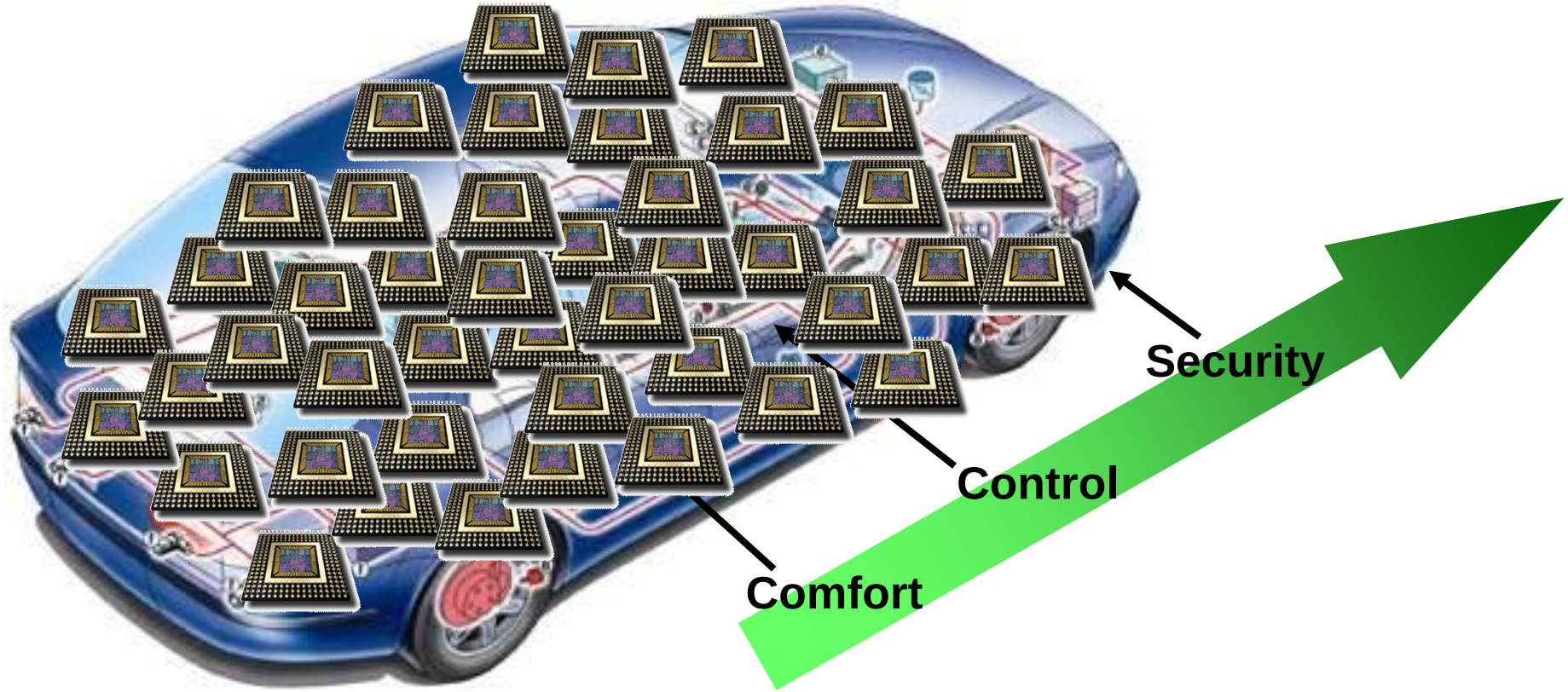







Systems are everywhere,
deeply involved in our daily life
and inter-connected.

Innovation = more functions = more computation

Some numbers…



→ *Tenth of interconnected processors*

→ *Hundreds of processing in parallel*

→ *Thousands of exchanged data*

GM Recalls 50,500 2011 Cadillac SRXs Over Airbag-related Software Glitch

By Robert Charette
Posted 22 Jun 2011 | 12:48 GMT



GM is recalling 50,500 Cadillac SRX c
of a software glitch that may not allow
airbags for passengers sitting in the rig
crash, reports a blog post at Zacks Inve

According to GM, the post says:

"...the front passenger

What Industry Needs from Architectu
nes: A Survey

Senior Member, IEEE, Henry
Antony Tang, Member, IE

The Standish Group Report

CHAOS



SYSTEM FAILURE

Released by:
Technical Operations
International Council on Systems Engineering (INCOSE)

Engineering Automotive Software

By the gigabyte of costly software that will be used in intelligent cars of the future, system and software development techniques and tools are required.

Manfred Broy, Ingolf H. Krüger, Alexander Pretschner, and Christian Salzmann

Abstract | The amount of software in cars grows exponentially. Driving forces of this development are the availability of cheaper and more powerful hardware, as well as the demand for innovation through new functionality. The rapidly growing significance of software and software-based functionality is the root of various challenges in the automotive industries. These concern their organization, definition of key competences, division of labor, logistics, maintenance, and long-term strategies. This paper pinpoints the driving forces of the domain, characterizes the essentials of automotive software, and discusses the challenges of automotive software engineering.

Keywords | Road vehicle electronics; software engineering; systems integration

I. Introduction

Just 30 years ago, the automotive industry witnessed the first deployment of tiny little bits of software in cars. It was to control the engine and, in particular, the ignition. Software-based solutions were strictly local, technically isolated, and did not relate to [illegible] independent and unconnected pieces [illegible] single dedicated controllers [illegible] A typical node ran a few [illegible] roughly a dozen



Telecom continues whirlwind of settlements with incorrect broadband meter readings

Oct. 19 (BusinessDesk) – Telecom has continued its rush to settle outstanding disputes ahead of next month's vote to split the company, reaching a deal with the Commerce Commission to repay broadband customers overcharged for incorrect meter readings.

The country's biggest phone company paid out $2.7 million to some 47,000 customers who were overcharged after a software glitch meant people hit their data limits early, the antitrust regulator said in a statement.

Telecom and the commission reached a settlement after the phone company acknowledged the fault and sought to compensate its customers. The regulator would waive its right to issue legal proceedings under the Fair Trading Act.

"We're pl

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Last updated at 21:44 GMT

rry users complain of fresh crash



C Radio 1 Newsbeat listeners described how the crash affected them

ackberry users have complained of a fresh crash hours after e company which makes the smartphones, RIM, said all rvices were "operating normally".

Twitter angry users reported renewed issues with their handsets d an inability to send messages and email.

e initial blackout saw Blackberry services across Europe, the ddle East and Africa disrupted - but that has now spread to Latin nerica.

M said the problems were caused by core and back-up switch lures.

e tweeter summed up the mood of many: "Blackberry server down AIN?!!! you have got to be kidding me!!!!!"

Related Stories

Can the iPhone still scare rivals?

Microsoft services h by failure

Android 'most popular' purchase

Why does it fail?

- Communication issues between numerous and various stakeholders.
- Time-to-market pressure vs. higher quality level.
- Ambiguous or incomplete descriptions of system.
- Non-availability of expertise for complex analysis.
- Manual-based methodologies.

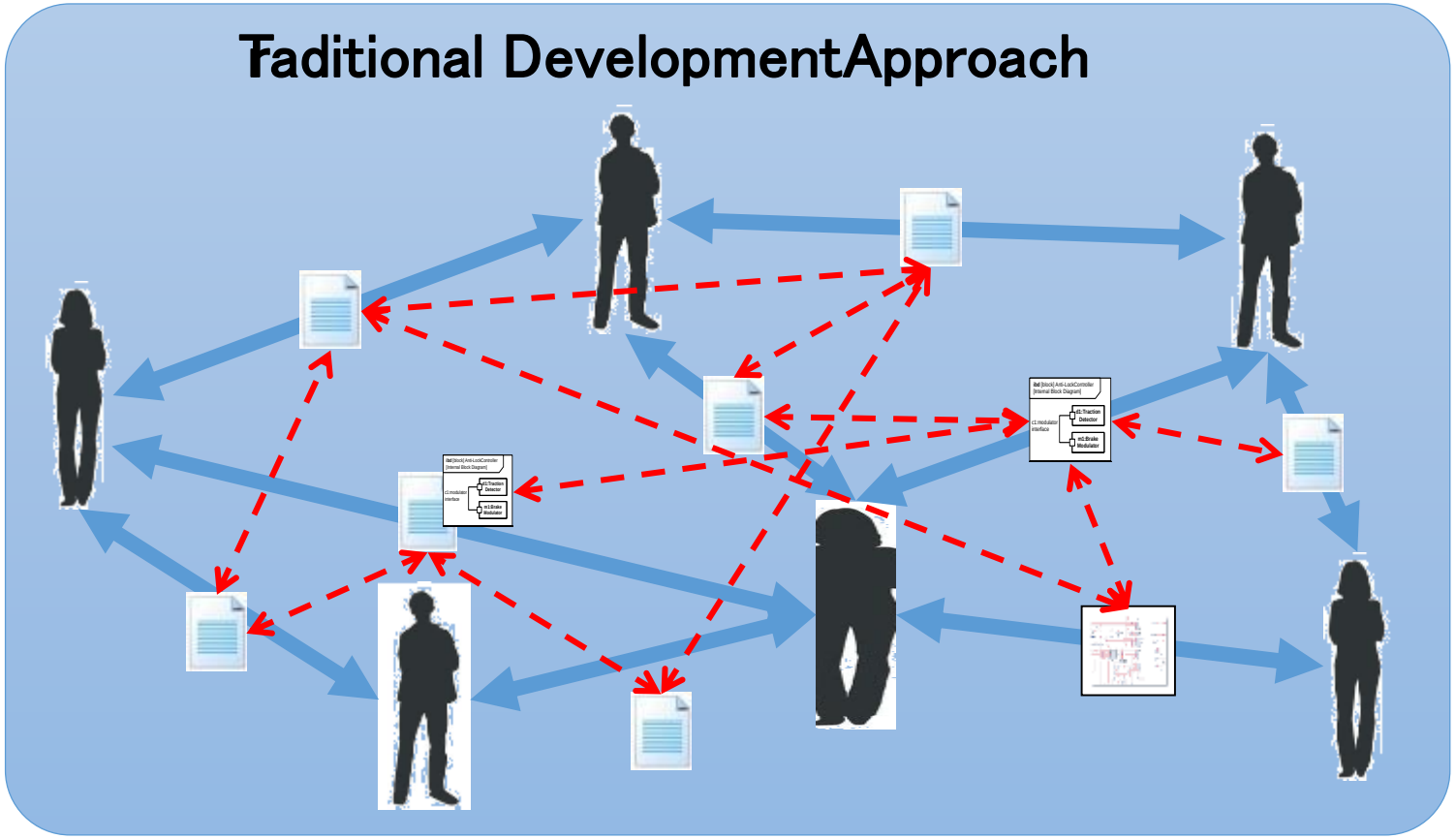
(Note: this list is of course not exhaustive)



Complexity, Complexity, **Complexity, Complexity, Complexity . . .**

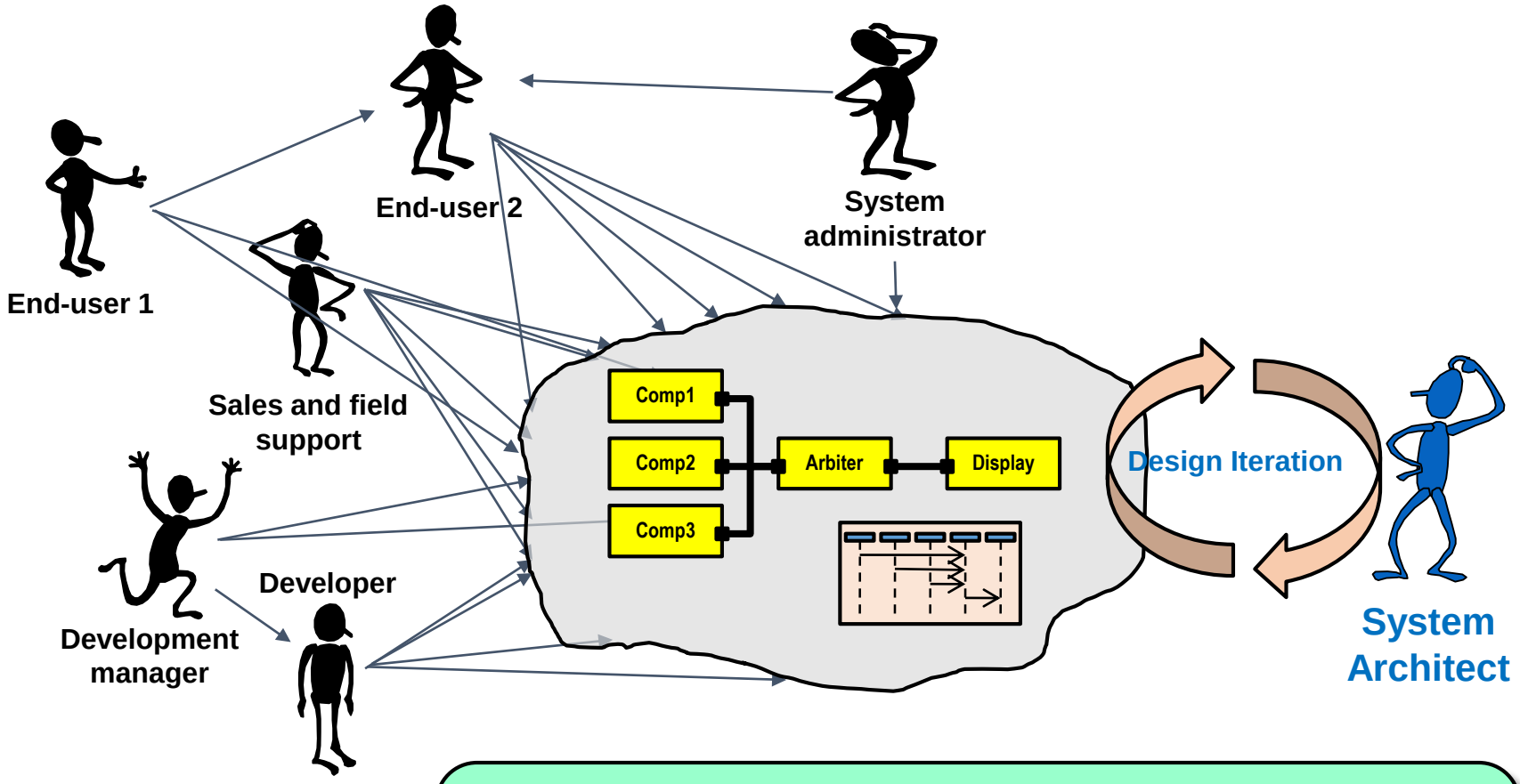


Problems of Traditional Approaches





How Architecture Helps Define Requirements



Many requirements conflicts and necessary tradeoffs are only detected through analysis of candidate architectures.

A System-Level Approach is Needed: Architecture!

Design the system as a whole rather than as an aggregate of separately designed sub-systems

- Provides possibility to ensure system integrity
- Requires a “big picture” approach

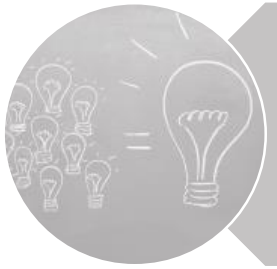
➔ An architecture specification

One definition of Architecture [IEEE Standard 1471]:

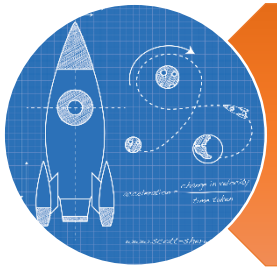
- “The fundamental organization of a system embodied in its components, their relationships to each other, and to the environment, and the principles guiding its design and evolution



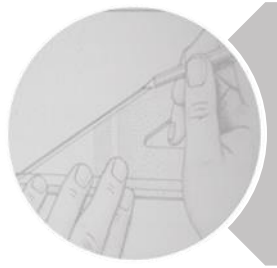
“To architect is to model”



Introduction



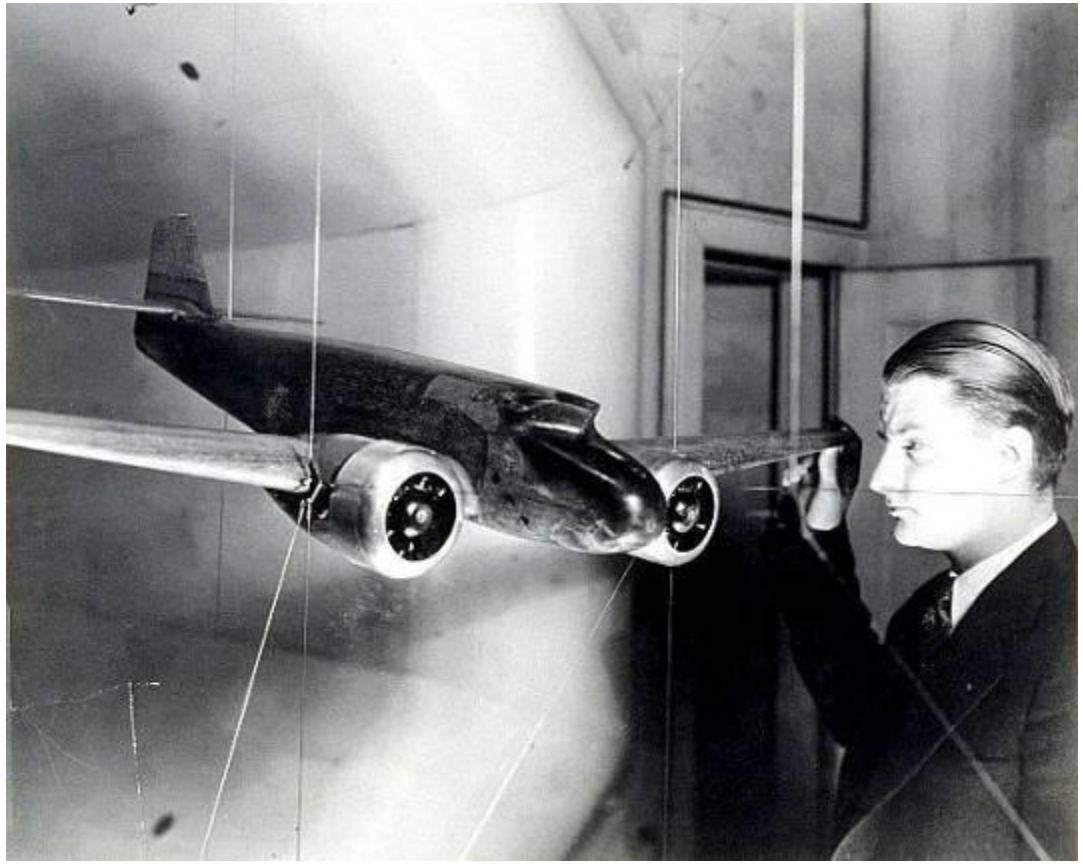
Basis is Modeling



Modeling with Which Language?

Models in Traditional Engineering

Probably as old as engineering



Extracted from B. Selic presentation

Engineering Models

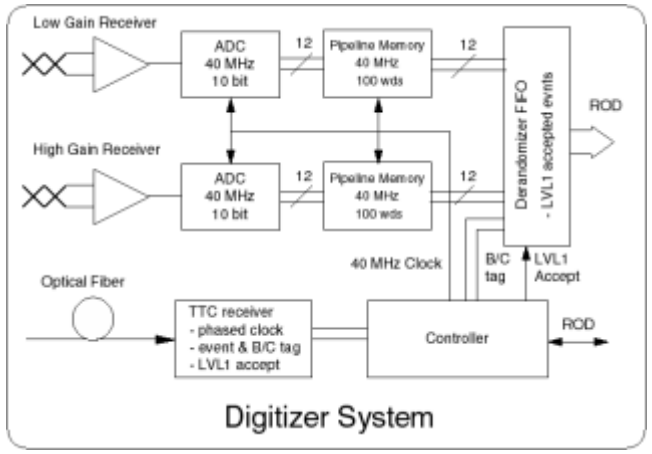
Extracted from B. Selic presentation

Engineering model

- A reduced representation of some system that highlights the properties of interest from a given viewpoint



Modeled system



Functional Model

- We don’t see everything at once
- We use a representation (notation) that is easily understood for the purpose on hand

Extracted from B. Selic presentation

Code is also an abstraction

```
SC_MODULE(producer)
{
    sc_outmaster<int> out1;
    sc_in<bool> start; // kick-start
    void generate_data () {
        for(int i =0; i <10; i++) {
            out1 = i ; //to invoke slave;
        }
    }
    SC_CTOR(producer) {
        SC_METHOD(generate_data);
        sensitive << start;
    }
}

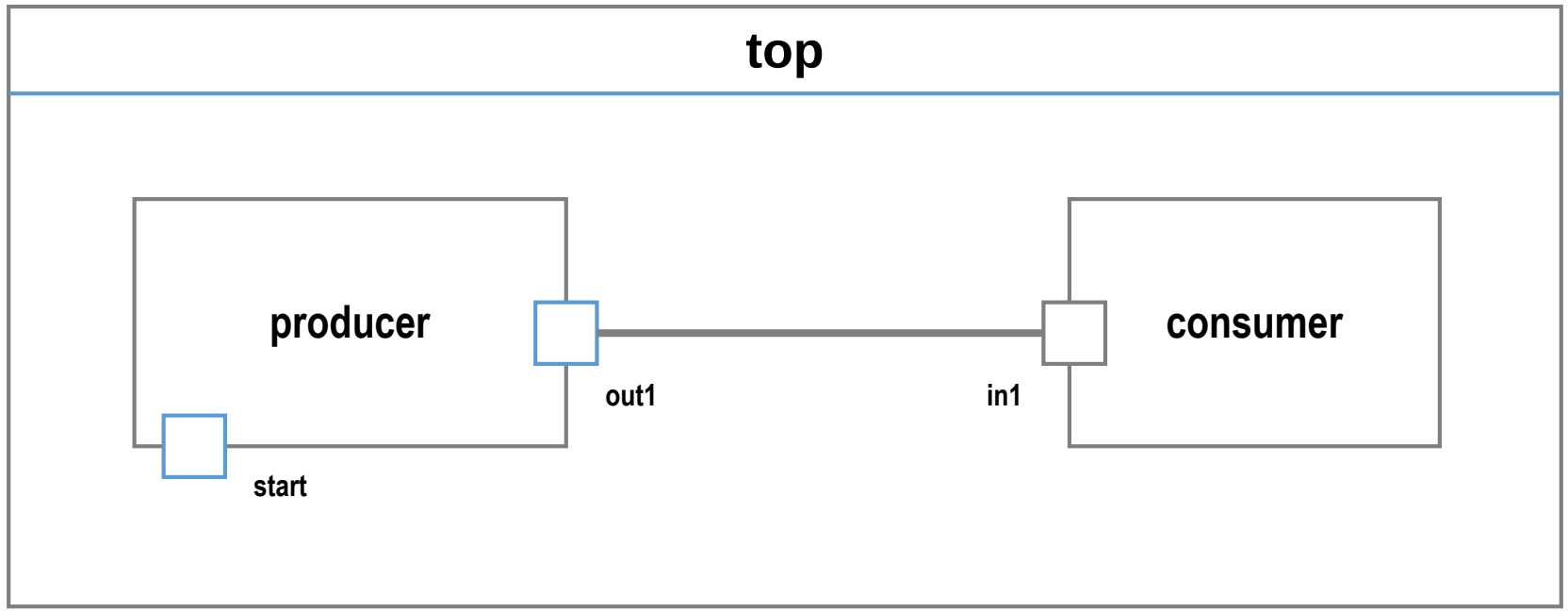
SC_MODULE(consumer)
{
    sc_inslave<int> in1;
    int sum; // state variable
    void accumulate (){
        sum += in1;
        cout << "Sum = " << sum << endl;
    }
    SC_CTOR(consumer)
    {
        SC_SLAVE(accumulate, in1);
        sum = 0; // initialize
    };
}

SC_MODULE(top) // container
{
    producer *A1;
    consumer *B1;
    sc_link_mp<int> link1;
    SC_CTOR(top)
    {
        A1 = new producer("A1");
        A1.out1(link1);
        B1 = new consumer("B1");
        B1.in1(link1);}};
```

Can you spot the architecture?

Extracted from B. Selic presentation

Architecture clarity

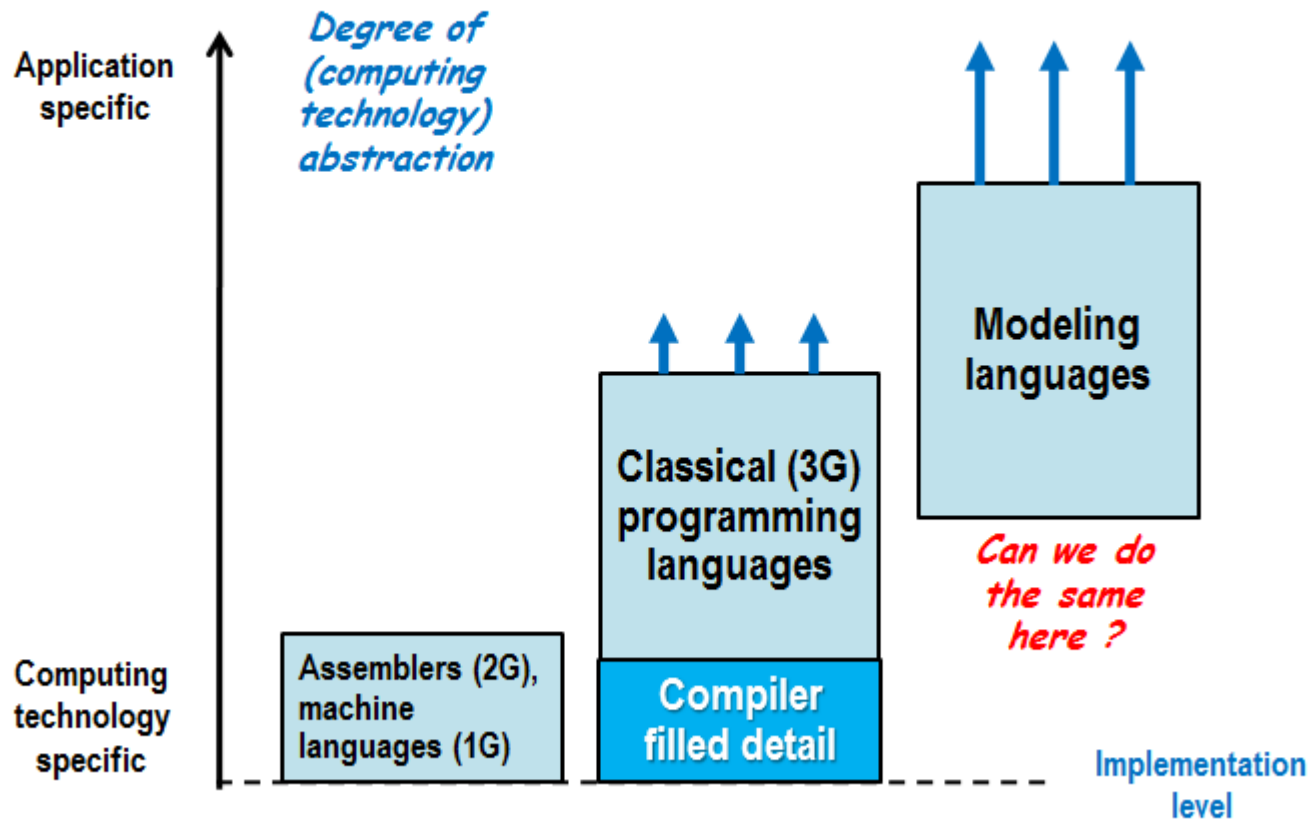


What about now?

Extracted from B. Selic presentation

The Evolution of Computer Languages

- Much of the evolution of computer languages is motivated by the need to be more human-centric (i.e., descriptive)



?

Is code a model?

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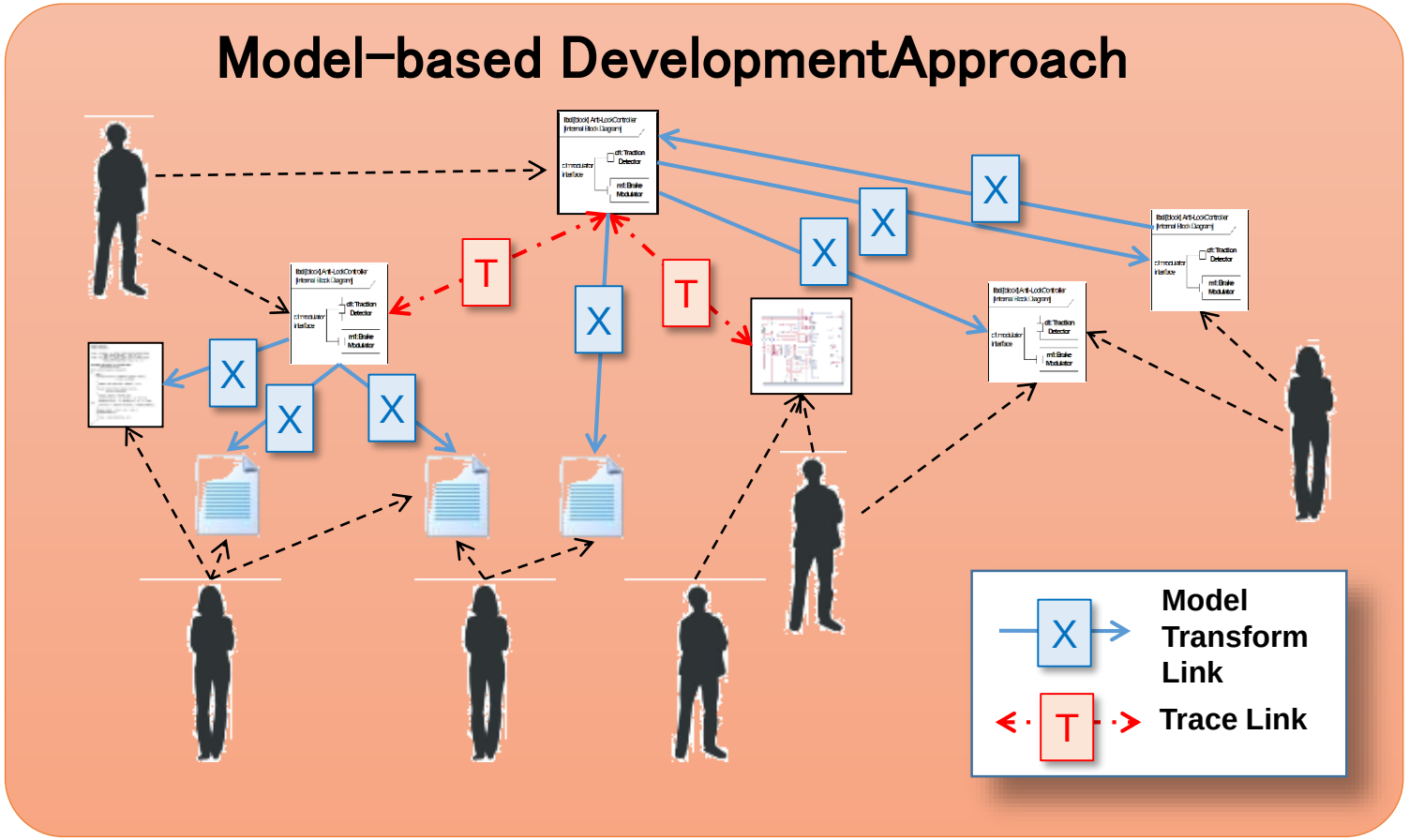
Two paradigms of MBE

Modeling Language



Computer-Aided Modeling and Engineering

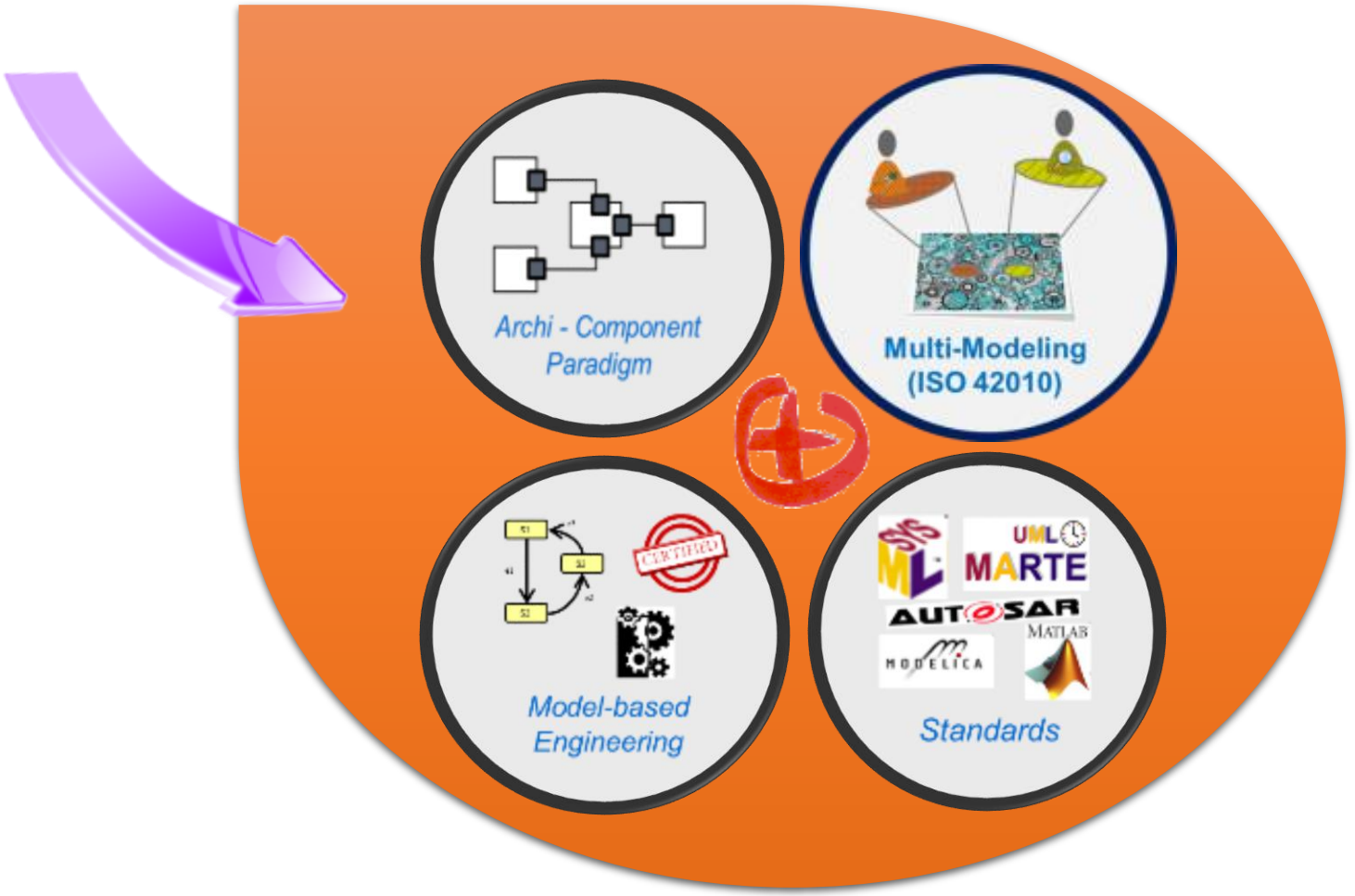
Why Model-Driven Engineering?





More efficient, More reliable,
and More scalable.

Going further for developing modern complex systems & software requires new advanced and innovative methods and tools

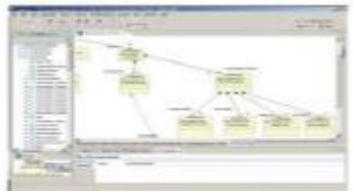


Numerous, Complex, and Interdependent Software

From requirement document: ***Hundreds of pages***





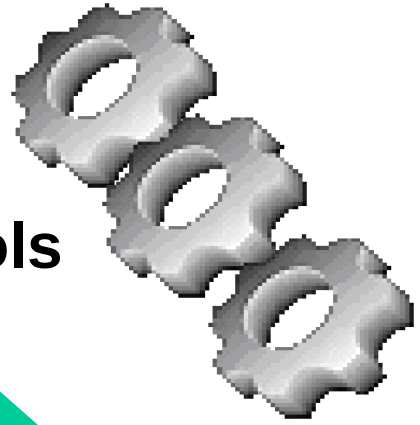


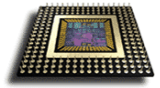






Methods **& tools**

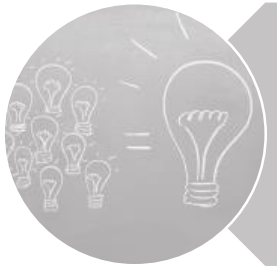




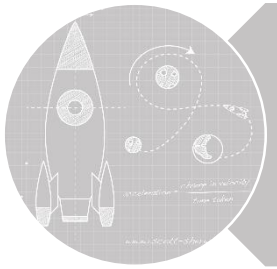




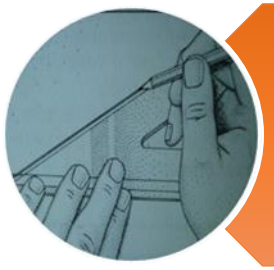
… to code: ***millions of lines***



Introduction



Basis is Modeling



Modeling with Which Language?

Building



Worker(s)



Architect

Computer science



Developer(s)

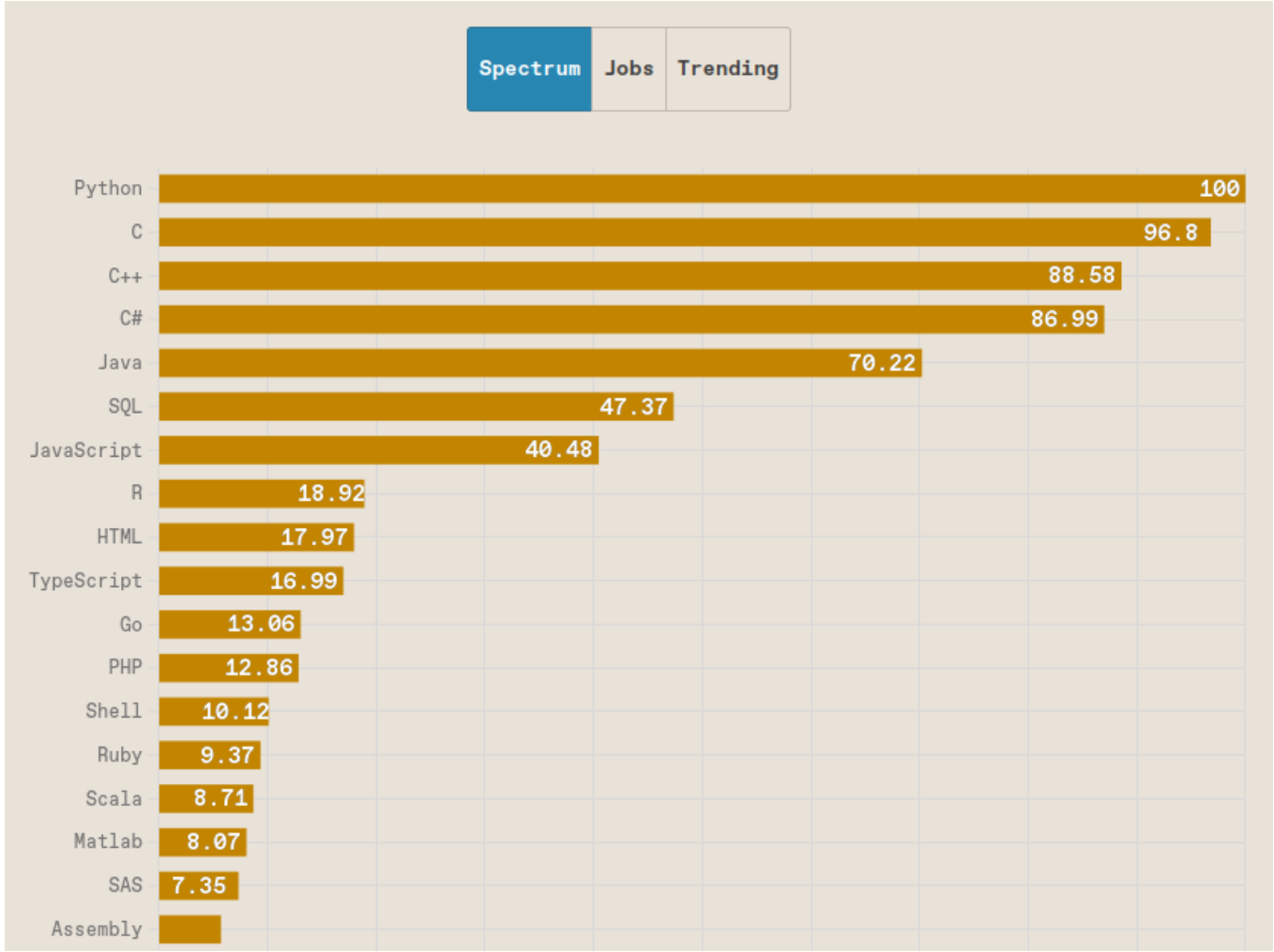


Designer, Architect

Popular programming languages

Language Rank	Types	Spectrum Ranking
1. Python		100.0
2. C		99.7
3. Java		99.5
4. C++		97.1
5. C#		87.7
6. R		87.7
7. JavaScript		85.6
8. PHP		81.2
9. Go		75.1
10. Swift		73.7

https://spectrum.ieee.org/computing/software/the-2017-top-programming-languages



https://spectrum.ieee.org/top-programming-languages-2022

And Now, What About Standards?

Standards have traditionally provided major boosts to technological progress!

But standards enable also vendor independence

- Users have a choice of different vendors (no vendor “tie-in”)
- Forces vendors into competing and improving their products

The Object Management Group (OMG) has created the Model-Driven Architecture initiative

- A comprehensive set of standards in support of MBE including standard modeling languages

UML – Unified Modeling Language



- Standardized by Object Management Group
- https://www.omg.org/spec/UML
- Current version 2.5.1
- Mainly targeted for SW development domain
- **Extensible** via profiles, for instance SysML or MARTE (realtime)

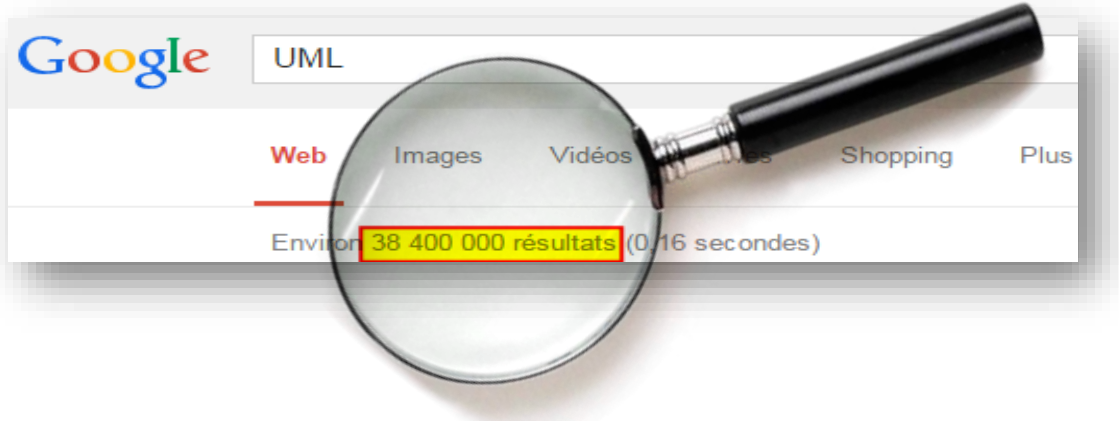






What is the Unified Modeling Language?

- Mature modeling language
  - Initially based on experienced modeling language designers: the three amigos, Booch, Jacobson and Rumbaugh but also Coleman, Desfray, Embley, Gamma, Harel, Meyer, Odell, Selic, Shaer-Mellor, Wirfs-Brock, etc…
  - 20 year old modeling language (current version 2.5.1), continually maintained and updated by experts from various origins: end users, tool providers and academics
  - Historically object-oriented software modeling language
- A rich modeling language covering a large set of concerns
  - E.g. architecture, automata, data-flow, scenario, and use-case
- Internationally popular and in-use
  - UML is widely educated, disseminated, and implemented… all around the world



UML, a standard modeling language

Standards have traditionally provided major boosts to technological progress!

- But standards enable also vendor independence
  - Users have a choice of different vendors (no vendor “tie-in”)
  - Forces vendors into competing and improving their products
- The Object Management Group (OMG) has created the Model-Driven Architecture initiative
  - A comprehensive set of standards in support of MBE including standard modeling languages, such as UML


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